

Francesco Cavina

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Quantitative data and AI engineer with a BSc in Econometrics and Data Science from Vrije Universiteit Amsterdam and MSc studies in Econometrics and Operations Research and Business Analytics. Experience in operational trading, Python and SQL workflows, forecasting pipelines, dashboards, model evaluation, data visualisation and AI ready data interfaces. Strong fit for Liquid Credit Desk work that combines market interest, quantitative analysis, data tooling and practical decision support.

TECHNICAL AND ANALYTICAL SKILLS

- **Quantitative analysis:** econometrics, probability, statistics, optimisation, forecasting, model evaluation, walk forward validation
- **Programming and data:** Python, SQL, PySpark, data ingestion, automated workflows, schema inference, data quality workflows
- **Data visualisation and decision support:** dashboards, performance visualisation, model signal comparison, assumption testing, self service tools
- **AI and data tools:** LLM workflows, RAG, ChromaDB, OpenAI API, Hugging Face, local LLM workflows, structured prompts
- **Engineering tools:** Git, Docker, FastAPI, Azure, Databricks, Airflow, Kubernetes, Terraform
- **Communication:** documentation, technical explanation, user focused prototypes, translating model output into practical recommendations

RELEVANT EXPERIENCE

Operational Trader

QuantFi
Netherlands

October 2022 - May 2023, Amsterdam,

- Developed and tested Python based trading strategies using Random Forest, XGBoost, neural networks and walk forward validation.
- Built data ingestion, retraining and daily forecast workflows in Python and SQL to support recurring trading decisions.
- Created dashboards and visualisations to compare model signals, realised performance and forecast quality.
- Evaluated model output with attention to overfitting, changing market conditions and practical usability.
- Worked in an environment where speed, clarity and disciplined decision making mattered daily.

Data & AI Platform Engineer

Robodata
Netherlands

September 2025 - June 2026, Amsterdam,

- Built Python, PySpark and SQL pipelines using Bronze, Silver and Gold data architecture.
- Developed metadata driven ingestion, schema inference and PII classification workflows for enterprise data platforms.
- Created local LLM workflows for data discovery, field classification and metadata generation.
- Built AI ready data interfaces that allowed governed datasets to be queried safely and consistently.
- Worked on self service tooling that made technical data infrastructure easier for users to understand and operate.
- Used Docker, Kubernetes, Terraform, Azure, Databricks, Airflow and SQLAlchemy in production platform work.

Teaching Assistant

Vrije Universiteit Amsterdam
Netherlands

June 2022 - November 2024, Amsterdam,

- Taught weekly Probability Theory tutorials to first year Econometrics students.
- Explained abstract mathematical concepts in a clear and practical way, with focus on reasoning rather than memorising procedures.
- Helped students understand the assumptions behind probability exercises and build confidence with formal problem solving.
- Built a student portfolio system with practical tools and information for first year students and represented the programme during Open Days.

EDUCATION

MSc in Econometrics and Operations Research

Track Business Engineering and Data Driven Decision Making • Vrije Universiteit Amsterdam

- Amsterdam, Netherlands • 2026-2029

MSc in Business Analytics

Track Computational Intelligence • Vrije Universiteit Amsterdam

- Amsterdam, Netherlands • 2026-2029

Bachelor of Science in Econometrics and Data Science

Vrije Universiteit Amsterdam • Amsterdam, Netherlands • 2021-2024 • 7.8/10 •

Thesis: Nonstandard Probabilities in an Internal Probability Space, grade 8.5/10

Minor in Pure Mathematics

University of Leeds • Leeds, England

- 2024 • 8.5/10

PROJECTS

- 1) [AI Powered Personal Website](https://github.com/FrancescoCavina02/My-Website) • <https://github.com/FrancescoCavina02/My-Website> • November 2025 - present
Next.js, TypeScript, FastAPI, Python, Tailwind CSS
Built a full stack portfolio with a FastAPI backend that parses an Obsidian vault into structured API endpoints for notes, quotes and chat functionality. Designed it as a self service knowledge interface, combining content modelling, backend automation, frontend usability and LLM based interaction.
- 2) [RAG Knowledge Chatbot](https://github.com/FrancescoCavina02/Spiritual-chatbot) • <https://github.com/FrancescoCavina02/Spiritual-chatbot> • November 2025 - present
Python, FastAPI, ChromaDB, sentence transformers, LLM APIs
Built a retrieval augmented chatbot using semantic chunking, vector search, structured prompts and citation based responses. Focused on reliable retrieval, traceable answers, hallucination reduction and reusable workflow design.
- 3) [Dodgeball Club Amsterdam](https://github.com/FrancescoCavina02/DCA-website) • <https://github.com/FrancescoCavina02/DCA-website> • June 2025 - present
Built and maintain a public website using HTML, CSS and JavaScript. Focused on clear navigation, simple content management and practical usability for club members.

Podcasts

[Podcast Link](#) - Created two independent interview podcasts with researchers in physics, cosmology, mathematics and philosophy.